

JavaScript Objects

Object

An Object is a collection of properties.

A property is an association between a name (or key) and a value.

For example, a person has a name, age, city, etc. These are the properties of the person.

1. Creating an Object

We can add properties into

`{}` as key: value pairs.

```
let person = {
```

```
  firstName: "Rahul",
```

```
  lastName: "Attuluri",
```

```
  age: 28,
```

```
};
```

```
console.log(person); // Object {firstName: "Rahul", lastName: "Attuluri", age: 28}
```

1.1 Identifiers

A valid Identifier should follow the below rules:

- It can contain alphanumeric characters, _ and \$.
- It cannot start with a number.

Valid Identifiers:

firstName;

\$firstName;

_firstName;

firstName12;

Invalid Identifiers:

12firstName;

firstName 12;

To use an Invalid identifier as a key, we have to specify it in quotes.

```
let person = {
```

```
  firstName: "Rahul",
```

```
  lastName: "Attuluri",
```

```
  age: 28,
```

```
  "1": "value1",
```

```
  "my choice": "value2",
```

```
};
```

```
console.log(person); // Object {1: "value1", firstName: "Rahul", lastName: "Attuluri",  
age: 28, my choice: "value2"}
```

2. Accessing Object Properties

2.1 Dot Notation

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
  "1": "value1",  
  "my choice": "value2",  
};
```

```
console.log(person.firstName); // Rahul  
Use Dot notation when the key is a valid Identifier.
```

2.2 Bracket Notation

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
  "1": "value1",  
  "my choice": "value2",  
};
```

```
console.log(person["firstName"]); // Rahul
```

2.3 Accessing Non-existent Properties

Dot Notation:

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
  "1": "value1",  
  "my choice": "value2",  
};
```

```
console.log(person.gender); // undefined
```

Bracket Notation:

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
  "1": "value1",  
  "my choice": "value2",  
};
```

```
console.log(person["gender"]); // undefined
```

2.4 Variable as a Key

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
};
```

```
let a = "firstName";
```

```
console.log(person[a]); // Rahul  
console.log(person.a); // undefined
```

2.5 Object Destructuring

To unpack properties from Objects, we use Object Destructuring. The variable name should match with the key of an object.

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,
```

```
};  
  
let { gender, age } = person;  
  
console.log(gender); // undefined  
  
console.log(age); // 28
```

3. Modifying Objects

3.1 Modifying Object Property

Dot Notation:

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
};  
  
person.firstName = "Abhi";  
  
console.log(person.firstName); // Abhi
```

Bracket Notation:

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
};  
  
person["firstName"] = "Abhi";  
  
console.log(person["firstName"]); // Abhi
```

3.2 Adding Object Property

Dot Notation:

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
};
```

```
person.gender = "Male";
```

```
console.log(person); // Object {firstName: "Rahul", lastName: "Attuluri", age: 28,  
gender: "Male"}
```

Bracket Notation:

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
};
```

```
person["gender"] = "Male";
```

```
console.log(person); // Object {firstName: "Rahul", lastName: "Attuluri", age: 28,  
gender: "Male"}
```

4. Property Value

The Value of Object Property can be

- Function
- Array
- Object

4.1 Function as a Value

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
  run: function () {  
    console.log("Start Running.");  
  },  
};
```

```
};
```

```
person.run(); // Start Running.
```

Methods:

A JavaScript method is a property containing a function definition.

For example, in

`document.createElement()`, the `document` is an Object, `createElement` is a key and `createElement()` is a Method.

4.2 Array as a Value

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
  habits: ["Playing Chess", "Singing"],  
};
```

```
console.log(person.habits); // ["Playing Chess", "Singing"]
```

```
console.log(person.habits[0]); // Playing Chess
```

```
console.log(person["habits"][1]); // Singing
```

4.3 Object as a Value

```
let person = {  
  firstName: "Rahul",  
  lastName: "Attuluri",  
  age: 28,  
  habits: ["Playing Chess", "Singing", "Dancing"],  
  car: {  
    name: "Audi",  
    model: "A6",  
    color: "White",  
  },  
};
```

```
};
```

```
console.log(person.car.name); // Audi
```

```
console.log(person.car["model"]); // A6
```